

Youssef Magdy

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[LinkedIn-Profile](#) | [GitHub-Profile](#)

Address: Tegel, Berlin

EDUCATION

- German International University in Berlin
Faculty of Media Engineering and Technology (graduate 2026) GPA: 1.46

WORK EXPERIENCE

- Junior Teaching Assistant for Data Structures and Algorithms with Java
o Help and explain to students in worksheets Sep.24 - Dec.24
- Software developer intern in Hotel Am Borsigturm, Germany Okt.23 - Nov.23
 - o Build Frontend website by using HTML & Tailwind
 - o Build Automated-PDF-Generator
- Internship as Technical Support by Orange Business Jul.24 - Aug.24
 - o Network Security, Technical Support, Cloud Partitioner and Cloud Solution Architect
- Cashier in my father's supermarket 2016 - 2023

TECHNICAL KNOWLEDGE

LLMs

Test Driven Development (TDD)

Neural Networks

Git & GitHub

Linux

Transfer Learning

OOP

RESTful APIs

Embedding

Fine-tuning

Agile, SCRUM, Jira

CNNs

PROJEKTS

1. [Face-Detector](#): A real-time face detection and tracking system built with **YOLOv3-face**, capable of detecting and capturing distinct faces from images, videos, or live webcam feeds.
 - Python – OpenCV library – Face_Recognition library – PyQt5
2. [E-Commerce Sentiment Analysis | Knowledge Graph | Graph-RAG-Chatbot](#): Intelligent E-Commerce Search powered by Knowledge Graphs + Large Language Models
 - Streamlit – Python – Neo4j Knowledge Graph – LangChain – HuggingFace LLMs
3. [Harris-SIFT-Detectors](#): A comprehensive analysis of corner detection and feature detection algorithms in computer vision, comparing **Harris Corner Detector** and **SIFT (Scale-Invariant Feature Transform)** techniques.
 - Python
4. [Lending-Club-Dataset-Data-Engineering](#): an **end-to-end data engineering and machine learning pipeline** for predicting **loan default status** using historical **LendingClub loan data (2007–2018)**.
 - SBERT Embedding – K-Means clustering – Feature Engineering – PowerBI
5. [LangChain-RAG-Streamlit](#): A **RAG** chatbot that allows users to upload a PDF and ask questions about its content.
 - LangChain – HuggingFace – Streamlit - **FAISS** for vector similarity search
6. [EuroSAT TransferLearning Classification](#): This project focuses on **Land Use and Land Cover (LULC) classification** using the **EuroSAT RGB dataset**. It compares a **baseline Convolutional Neural Network (CNN)** trained from scratch with a **transfer learning approach using EfficientNetB0** pretrained on ImageNet
 - Transfer Learning – Tensorflow – CNN – EfficientNetB0

7. [gpt2-124M-parameters](#): a PyTorch implementation of a GPT-2-style Transformer model, built from scratch by following the core ideas from the "Attention is All You Need" paper.
 - Deep Neural Networks – Transformer Architecture – LLM – Embedding
8. [Smart-Irrigation-Predictor Bachelor Project \(DNN & RAG system\)](#): A machine learning-powered web application that predicts **Evapotranspiration (ET)** using 15 years of historical and forecasted weather data. It offers intelligent irrigation recommendations tailored to location, crop, and soil type — supporting **sustainable agriculture** through data-driven decisions.
 - Deep Neural Networks – Tensorflow – RAG – Vercel Frontend Deploy – Render Backend Deploy – API
9. [Music-Generator-Model Deep Neural Network](#): A Deep Neural Network trained on irish.abc file that contains 817 songs and generates likely. [Watch demo video](#)
 - Deep Neural Networks – Tensorflow
10. **AWS EC2 instances**: we were downloading 60000 books to a website, so we had to rent instances (computers)
 - Linux – AWS – Bash
11. [Names-Generator-Model Deep Neural Network](#): This project demonstrates the evolution of name generation using progressively more powerful neural network models that trained on 17000 names.
 - Deep Neural Networks – PyTorch – Embedding
12. [Full-Stack-Website \(Group projekt\) with MERN-Stack](#): Teer Enta brings together personalized recommendations, seamless booking, and smart budgeting into one intuitive platform.
 - MongoDB – Express.js – React.js – Node.js – Tailwind – Agile – SCRUM – Jira – Restful APIs
13. [Charity web application](#): building a website that Grants people the ability to connect with non-profit organizations in a much easier form so that they can donate items needed to help people in need, and today we want to showcase that project with you all: Introducing “Sharelkhair”, your platform that connects Donors to People in need like never before!
 - React – Tailwind
14. [Implementation of Tomasulo-Algorithm](#): implementing Tomasulo algorithm that manages how the sequential operations run efficiently, act like a compiler.
 - Java – Javafx – OOP
15. [Full Chess Game](#): implemented Chess game
 - Java – Javafx – OOP
16. [Automatik PDF-Generator](#): algorithm that takes an excel file that contains customer and generates PDF for every customer, used for **Hotel am Borsigturm in Germany**.
 - Python – Test Driven Development (TDD)
17. **Health system**: built Entity Relationship diagram of a hospital structure by using SQL and as frontend ASP.NET.
 - SQL – ASP.NET
18. [Fillet-O-Neumann Computer architecture](#): The Von Neumann architecture is a computer design where both program instructions and data are stored in the same memory unit, using the same pathways or bus system
 - C
19. [Operating-system-Scheduler](#): Scheduler manages the usage of resources between the different processes. Mutexes that ensure mutual exclusion over the critical resources. Scheduler that schedules the processes that we have in our system.
 - C
20. [islamic classroom](#): like Google classroom, made for a small school that teaches children about Islamic literature, that reminds the children’s parents for doing homework or attendance or posts and so on.
 - Flutter – Firebase – Supabase – APIs

LANGUAGES

Programming languages: Java | Python | C | C++ | SQL | HTML | CSS | Javascript

Frameworks: PyTorch | Tensorflow | Scikit-learn | LangChain | React.js | Node.js | Express.js | MongoDB | Tailwind | ASP.NET

Communications languages: German (fluent) | English (fluent)

CONTACTS

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